BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 6485

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry - Paper: I - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6485

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. The change in heat energy of a chemical reaction at a constant temperature and pressure is called:

(A) Enthalpy change
(B) Bond energy
(C) Heat of sublimation
(D) Internal energy

Q2. For which system, does the equilibrium constant Kc has units of (concentration)$^{-1}$?

(A) $N2(g) + 3H2(g) \rightleftharpoons 2NH3(g)$
(B) $H2(g) + I2(g) \rightleftharpoons 2HI(g)$
(C) $2NO2(g) \rightleftharpoons N2O4(g)$
(D) $2HF(g) \rightleftharpoons H2(g) + F2(g)$

Q3. Colligative properties are the properties of:

(A) Dilute solutions which behave as nearly ideal solutions
(B) Concentrated solutions which behave as nearly non-ideal solutions
(C) Both A and B
(D) Neither A nor B

Q4. The cathodic reaction in the electrolysis of dil.H_2SO_4 with Pt electrode is:

(A) Reduction
(B) Oxidation
(C) Both oxidation and reduction
(D) Neither oxidation nor reduction

Q5. The molar volume of CO_2 is maximum at:

(A) STP
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2 atm

Q6. Which of the given pair do not show isomorphism?

(A) $NaNO_3$, KNO_3
(B) $ZnSO_4$, $NiSO_4$
(C) Cu, Ag
(D) NaCl, $CuCl_2$

Q7. Which one of the given compounds possess ionic bonding?

(A) CaO
(B) CH_4
(C) CH_3Cl
(D) C_2H_6

Q8. Catalyst for a catalyst is also called:

(A) Promotor
(B) Inhibitor
(C) Poisoning
(D) Retarder

Q9. The mass of one mole of electrons is:

(A) 1.008 mg
(B) 0.55 mg
(C) 0.184 mg
(D) 1.673 mg

Q10. Which of the given sub-atomic particle does not show ionization?

(A) Electron
(B) Proton
(C) Neutron
(D) Alpha ray

Q11. The type of hybridization in molecule of ethene ($CH_2 = CH_2$) is:

(A) sp
(B) sp^3
(C) sp^2
(D) dsp

Q12. Quantum number values for 2p orbitals are:

(A) n=2, l=1
(B) n=1, l=2
(C) n=1, l=0
(D) n=2, l=0

Q13. Acetone and chloroform are soluble into each other due to:

(A) Intermolecular hydrogen bonding
(B) Ion dipole interaction
(C) Instantaneous dipole
(D) All of these

Q14. Which of the given will have same number of molecules at STP?

(A) $280cm^3$ of CO_2 and $280cm^3$ of N_2O
(B) $11.2dm^3$ of O_2 and 32g of O_2
(C) 44g of CO_2 and 11.2 dm^3 of CO_2
(D) 28g of N_2 and 5.6 dm^3 of oxygen

Q15. During the process of crystallization, the hot saturated solution is:

(A) Cooled very slowly to get large-sized crystals

(B) Cooled at a moderate rate to get medium sized crystals

(C) Evaporated to get the crystals of the product

(D) Mixed with an immiscible liquid to get the pure crystals of the product

Q16. The comparative rates at which the solutes move in paper chromatography, depends on:

(A) The size of paper

(B) Rf Values of salutes

(C) Temperature of the experiment

(D) Size of the chromatographic tank used

Q17. 27g of Al will react completely with how much mass of O_2 to produce Al_2O_3.

(A) 8g of oxygen

(B) 16g of oxygen

(C) 32g of oxygen

(D) 24g of oxygen

Chemistry - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note:- Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Define Relative Atomic Mass also give two examples.

ii. How is the law of conservation of mass obeyed during stoichiometric calculations?

iii. Why do the isotopes have same chemical but different physical properties?

iv. State Distribution Law.

v. How are coloured impurities removed from crystals?

vi. Write two uses of chromatography.

vii. Prove that d = PM/RT

viii. Calculate the value of R in SI units.

ix. Give two applications of plasma.

x. What are buffers?

xi. What is the effect of common ion on solubility?

xii. How Kc determines the direction of chemical reaction?

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Evaporation takes place at all temperatures. Give reason.

ii. Iodine dissolves readily in tetrachloromethane. Give reason.

iii. Define Transition Temperature. Give an example.

iv. The electrical conductivity of the metals decreases by increasing temperature. Why?

v. Why is e/m value of cathode rays just equal to that of electron?

vi. State Aufbau Principle.

vii. State Heisenberg's uncertainty principle. Give its mathematical form.

viii. Cathode rays are material particles. Justify it.

ix. The sum of mole fractions of all the components is always equal to unity for any solution. Justify it.

x. What are conjugate solutions? Give an example.

xi. What is rate determining step? Give an example.

xii. Write two applications of enzyme catalysis.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. Why Helium can not exist as diatomic molecule?

ii. Draw out Lewis structures of (i) BF_3 (ii) CH_4

iii. The distinction between co-ordinate covalent bond and a covalent bond vanishes after bond formation in NH_4^+, explain.

iv. Why the dipole moment of CO_2 is zero?

v. Define Standard Enthalpy of Formation with an example.

vi. Is it true, non-spontaneous process never happens in universe?

vii. Why burning of candle is spontaneous process?

viii. Find out oxidation state of Mn in $KMnO_4$.

ix. Why a salt bridge maintains electrical neutrality in the cell?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 8) (a) What is yield? Write its types. How will you calculate the percentage yield? (1+1+2) (b) What are quantum numbers? Give importance of azimuthal quantum number. (1+3)

Q6.. (Marks 8) (a) A gas having a volume of $10dm^3$ is enclosed in a vessel at 0°C and the pressure is 2.5 atm. This gas is allowed to expand until the new pressure is 2

atm. What will be the new volume of this gas, if the temperature is maintained at 273K? (b) Write down four applications of electrochemical series.

Q7.. (Marks 8) (a) Define hybridization and explain sp^2 hybridization by giving the example of ethene ($CH_2 = CH_2$) (b) How the enthalpy of a reaction can be measured by using bomb calorimeter.

Q8.. (Marks 8) (a) What is hydrogen bonding? Explain H-bonding in biological compounds. (b) The solubility of CaF_2 in water at 25°C is found to be 2.05×10^{-4} mole dm^{-3}. What is the value of Ksp at this temperature?

Q9.. (Marks 8) (a) What is Roult's law? Explain it by three ways. (b) Define Order of Reaction. Give examples of some reactions explaining the order of reaction.

GROUP – II

Chemistry - Paper: I - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6484

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. When 6d orbital is complete, the entering electron goes into:

(A) 7f
(B) 7s
(C) 7p
(D) 7d

Q2. Acetone and chloroform are soluble in each other due to:

(A) Intermolecular hydrogen bonding
(B) Instantaneous dipole
(C) Ion dipole interaction
(D) Dipole-dipole forces

Q3. The deviation of a gas from ideal behaviour is maximum at:

(A) -10°C and 5.0 atm
(B) -10°C and 2.0 atm
(C) 100°C and 2.0 atm
(D) 0°C and 2.0 atm

Q4. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of total pressure exerted by oxygen is:

(A) 1/3
(B) 8/9
(C) 1/9
(D) 16/17

Q5. When water freezes at 0°C, its density decreases due to:

(A) Cubic structure of ice
(B) Empty spaces present in the structure of ice
(C) Change of bond lengths
(D) Change of bond angles

Q6. Solvent extraction method is a particularly useful technique for separation when the product to be separated is:

(A) Non-volatile or thermally unstable
(B) Volatile or thermally stable
(C) Non-volatile or thermally stable
(D) Volatile or thermally unstable

Q7. The drying agent used in vacuum desiccator is:

(A) Silica gel
(B) NaCl
(C) AgCl
(D) I_2

Q8. The number of isotopes of Na are:

(A) 9
(B) 11
(C) 6
(D) 3

Q9. The largest number of molecules are present in:

(A) 3.6g of H_2O
(B) 4.8g of C_2H_5OH
(C) 2.8g of CO
(D) 5.4g of N_2O_4

Q10. If a strip of Cu metal is placed in a solution of $FeSO_4$:

(A) Cu will be deposited
(B) Fe is precipitated out
(C) Cu and Fe both dissolve
(D) No reaction takes place

Q11. The rate of reaction:

(A) Increases as the reaction proceeds
(B) Decreases as the reaction proceeds
(C) Remains the same as the reaction proceeds
(D) May decrease or increase as the reaction proceeds

Q12. 18g glucose is dissolved in 90g of water. The relative lowering of vapour pressure is equal to:

(A) 1/5
(B) 5.1
(C) 1/51
(D) 6

Q13. An excess of aqueous $AgNO_3$ is added to aqueous $BaCl_2$ and precipitate is removed by filtration. What are the main ions in the filtrate?

(A) Ag^+ and NO_3^- only
(B) Ba^{2+} and NO_3^- only
(C) Ag^+, Ba^{2+} and NO_3^-
(D) Ba^{2+}, NO_3^- and Cl^-

Q14. The net heat change in a chemical reaction is same, whether it is brought about in two or more different ways in one or several steps. It is known as:

(A) Henry's Law
(B) Hess's Law
(C) Joule's Principle
(D) Law of Conservation of Energy

Q15. Which of the given species has unpaired electrons in anti-bonding molecular orbitals?

(A) [illegible in source scan]
(B) N_2^+
(C) B_2
(D) F_2

Q16. The number of bonds in nitrogen molecule is:

(A) One σ and one π
(B) One σ and two π
(C) Three sigma only
(D) Two sigma and one π

Q17. In the ground state of an atom, the electron is present:

(A) In the nucleus
(B) In the second shell
(C) Nearest to the nucleus
(D) Farthest from the nucleus

Chemistry - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note:- Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8x2 = 16)

i. Law of conservation of mass has to be obeyed during stoichiometric calculations. Justify.

ii. What do you mean by molar volume? Give an example.

iii. What is empirical formula? Give an example.

iv. Define Solvent Extraction and Partition Law.

v. Differentiate between stationary and mobile phase.

vi. Define Sublimate. Give two examples.

vii. Explain that the process of respiration obeys the Dalton's law of partial pressure.

viii. What is Avogadro's law of gases?

ix. Derive the value of R when the pressure is in Nm^{-2} and volume in m^3.

x. What is the pH of a solution? Write formula to calculate pH of a solution.

xi. Write the effect of common ion on solubility. Give an example.

xii. Define Lowery Bronsted Concept of Acid and Base.

Q3.. (Marks 16) Write short answers to any Eight parts. (8x2 = 16)

i. What are dipole-dipole forces? Give example.

ii. Why ethane (C_2H_6) has higher boiling point than methane (CH_4).

iii. How fish and plants survive under ice for months in winter?

iv. Why evaporation causes cooling?

v. Give two properties of cathode rays.

vi. How positive rays are produced?

vii. What are slow and fast neutrons?

viii. Why Rutherford's model failed?

ix. Define Molality. Give its equation.

x. Differentiate between ideal and non-ideal solution in two aspects.

xi. Give two characteristics of enzyme catalysis.

xii. Define Auto Catalyst with an example.

Q4.. (Marks 12) Write short answers to any Six parts. (6x2 = 12)

i. Cationic radius is smaller than that of its parent atomic radius, why?

ii. Explain geometry of H_2S molecule on the basis of VSEPR theory.

iii. Define State and State Function.

iv. How electronegativity changes in a group?

v. Define Co-ordinate covalent bond with a suitable example.

vi. Burning of a candle is a spontaneous process, justify.

vii. What do you mean by enthalpy of neutralization? Give a suitable example.

viii. What is the function of salt bridge?

ix. Define Standard Electrode Potential and how it is represented?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 8) (a) Write a note on limiting reactant and explain by giving two examples. (2+2) (b) Write down postulates of Bohr's atomic model. (4)

Q6.. (Marks 8) (a) Describe four applications of electrolysis processes of industrial importance. (4) (b) A gas having a volume of 10dm^3 is enclosed in a vessel at 0°C

and the pressure is 2.5 atm. This gas is allowed to expand until the new pressure is 2atm. What will be the new volume of this gas, if the temperature is maintained at 273K? (4)

Q7.. (Marks 8) (a) How the enthalpy of combustion is measured out by bomb calorimeter? (4) (b) Draw out geometry of O_2, N_2 according to M.O.T. (4)

Q8.. (Marks 8) (a) What are liquid crystals? Give their three uses. (2+3) (b) The solubility of PbF_2 at 25°C is 0.64g dm^{-3}. Calculate Ksp of PbF_2. (4)

Q9.. (Marks 8) (a) Explain the three statements of Raoult's law. (4) (b) How does Arrhenius equation help us to calculate the energy of activation of a reaction? (4)